Literature Review: Explanation Using Defeasible Rule Sets

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Abstract

Explainable AI seeks to develop models whose decisions are not only accurate but align with human-understandable reasoning. While traditional rule-based classifiers like CORELS offer transparency, they often struggle to capture the "normally... unless..." structure of defaults and exceptions that reflect how humans think. This paper investigates the Exception Closed Conjunctive Rule Sets (ECCRS) framework, a symbolic classifier that embeds exception handling directly into its rule structure. Our primary contribution is a verification workbench that translates ECCRS rulesets into Kraus, Lehmann, and Magidor (KLM) conditional knowledge bases. We evaluate the alignment between Most Specific Wins (MSW) predictions and formal defeasible entailment under three distinct closure methods: Rational Closure, Lexicographic Closure, and System W. This project compares the two explanation traces produced by ECCRS and the different closure algorithms when applied to the KLM conditional knowledge base. By comparing these traces, we demonstrate that the human-readable justifications generated by the symbolic classifier are consistent with the conclusions derived through formal defeasible reasoning methods.

1 Introduction

When a model denies a loan or flags a patient as high-risk, citing a confidence score is not a satisfactory explanation. Machine learning models are increasingly prevalent in areas where decisions have a direct impact on individuals, such as healthcare, finance, and justice. In these areas, users need to understand not just what a model decided, but how exactly it came to that conclusion. Post-hoc explanations, such as surrogate models, offer insight into black-box behavior, but they cannot guarantee that the reasoning they reveal matches the model's internal logic. This fundamental limitation motivates the use of fully transparent symbolic classifiers.

Symbolic rule-based models represent logic explicitly as sequences of if-then rules. This allows every decision to be traced and explained. Existing approaches, including CORELS rule lists and Boolean formula classifiers, achieve high predictive accuracy while maintaining transparency, but they fail to capture the structure of human reasoning involving defaults and exceptions. Exceptions must be encoded as separate rules or nested formulas, which can obscure their relationship to the rules they override.

Madzime et al. [14] address this gap with the Defeasible Horn Classifier with Exceptions (DHCE), which combines defaults and exceptions into normally-unless statements, producing explanations that align with human reasoning. It provides traces showing which defaults apply and which exceptions override them, but the framework is informal and cannot guarantee correctness. ECCRS extends DHCE by embedding exception handling in a formal non-monotonic framework, ensuring that predictions are both transparent and verifiable. By translating ECCRS rulesets into KLM conditional knowledge bases and applying closure methods, we allow both formal verification and human-readable explanations. The explanation traces are the mechanism that connects these two sides, making the verification legible rather than purely formal.

This review starts with comparing post-hoc explanations to symbolic classifiers that are transparent by design. We examine why earlier approaches struggle to handle the defaults and exceptions that humans use naturally, which motivates our focus on the ECCRS framework as a way to bridge intuitive explanation and formal proof. We introduce the technical structure of ECCRS and its Most Specific Wins operator before moving onto the formal foundations of the KLM framework. We analyze the trade-offs between Rational Closure, Lexicographic Closure, and System W. We then explain the translation of ECCRS rulesets into KLM conditional knowledge bases. By establishing this link, we can then use two explanation traces to verify that the human-readable justifications generated by the symbolic classifier are enforced by defeasible logic entailment.

2 Explainable AI and Symbolic Learning

2.1 Interpretability in Machine Learning

Machine learning models are being used in increasingly consequential areas of our lives, specifically in areas where decisions can have a noticeable impact on the individuals involved, for example, medical diagnoses and loan applications. It is a natural response to want to know the reasoning behind these decisions. Gilpin et al. [6] highlight an important difference between interpretability, which is understanding what a model did, and explainability, which goes further and requires that a model be able to justify the process behind its reasoning. The point is raised that, while explainable AIs are always interpretable, it is not the case that interpretable models are always explainable.

An example of this is COMPAS, a criminal risk assessment tool used in the United States, which produced conclusions that were racially biased and unreliable [6, 18]. Cases like this have led to more attention being paid to regulations. Doshi-Velez and Kim [5] have argued that the EU now requires that automated systems, including AI models, explain decisions that have an impact on users. They also argue that interpretability is one of the most practical ways to ensure fairness and safety in a system.

There are two ways of solving the problem. The first involves explaining black-box models afterwards (post-hoc) using a simpler, "surrogate" model. Ribeiro et al. [16] do this using their LIME framework. LIME works by approximating the black-box model locally using a simpler interpretable model which is trained on perturbed versions of the input. The second is to use models that are already completely transparent. Rudin [18] advocates that the second method is superior, arguing that the simpler model used to represent a black box is not the same as the black box itself

and can lead to incorrect explanations being generated. In other words, for important decisions, a post-hoc explanation is not a valid replacement for interpretability. This motivates our turn towards symbolic rule-based classifiers, which will be discussed in the next section.

2.2 Symbolic Learners and Rule-Based Models

We now introduce symbolic rule-based classifiers. These models represent their logic explicitly, in the form of if-then rules that we are able to read and follow. Symbolic classifiers make it possible to trace how we reach a particular conclusion.

Before introducing rule-based classifiers, we note that even symbolic classifiers require formal explanation methods, which is what Darwiche and Hirth [4] provide the basis for. They present an approach to symbolic explanation. They introduce the idea of a complete reason, which is a necessary and sufficient condition for why a decision was made. From this, shorter sufficient reasons, called abductive or PI-explanations, can be derived. This then helps us to explain that the decisions are correct through the use of a logic-based method, guaranteeing correctness.

Lists of rules are one of the most natural symbolic classifiers. Angelino et al. [1] use their CORELS algorithm to show that it is possible to learn certifiably optimal rule lists, which are sequences of ordered if-then rules where predictions are made by the first appearing instance of a matching rule. Applied to the same COMPAS dataset discussed in Section 2.1, CORELS produces a list of rules that is transparent and also directly matches the accuracy of the black-box system. In this setting, this challenges the assumption that by prioritising interpretability, we sacrifice performance and quality of predictions. Jaakkola et al. [7] take a similar approach, but using short Boolean formulas as classifiers. They find that cross-validation limits formula length, since longer formulas begin to overfit before they become significantly more accurate. Importantly, this length limit is not just a technical constraint. Shorter formulas are also more interpretable, meaning that avoiding overfitting and improving readability come hand in hand.

These approaches are transparent, but share a common problem. They don't capture the form of reasoning that makes up much of human decision-making, a default and exception type of reasoning. In a list of rules, an exception to a default must be defined as a distinct, higher-priority rule. This breaks the natural connection between the two. In Boolean formulae, exceptions to defaults are located in often complicated nested logical structures without any clear link to the defaults they override. Rienstra [17] introduces a problem with symbolic explanation approaches that ignore background knowledge about the data. Rienstra says that sufficient reasons, like those of Darwiche and Hirth [4] can be overly long because they include features that are technically relevant but redundant given what is known about normal feature relationships in the data. Rienstra proposes extending the explanation system using non-monotonic formalisms to help solve this problem.

Madzime et al. [14] introduce the Defeasible Horn Classifier with Exceptions (DHCE) as a way of addressing the default-and-exception gap. DHCE learns ranked Horn rules in which each default is grouped together with its exceptions, producing a "normally... unless..." statement that reflects how humans make decisions. Each prediction generates a step-by-step trace showing which default applied and which exceptions then overrode it, without using any post-hoc tools. Evaluated on standard binary classification benchmarks, DHCE either matches or outperforms other interpretable models. However, DHCE isn't based in a formally defined non-monotonic reasoning framework, which means it lacks the guarantees that a known framework would provide. This limitation is what motivates ECCRS, which addresses the same design goals but places them in a formal non-monotonic framework, providing the theoretical guarantees that come with using a formal framework.

2.3 ECCRS as a Symbolic Classifier

Exception Closed Conjunctive Rule Sets (ECCRS) is a symbolic rule-based classifier designed to help solve the limitations of existing interpretable approaches [13]. Unlike decision lists, which draw conclusions using ordered rules, or Boolean formulas, which handle exceptions in nested logical structures, ECCRS includes the relationship between defaults and exceptions in the structure of the rules themselves. It focuses on the same limitations as DHCE [14] but uses a formal non-monotonic framework to refine the approach.

An ECCRS ruleset consists of rules of the form (B, ℓ), where B is a consistent conjunction of signed literals and $\ell \in \{0, 1\}$ is a class label. A rule (B, ℓ) is applicable to an instance F if $F \models B$. The key structural constraint is that when two rules can both apply to the same instance, one must be strictly more specific than the other. Formally, for rules (B, ℓ) and $(C, 1-\ell)$, we require either $B \subset C$ or $C \subset B$, so that one rule is strictly more specific than the other. This is the exception closure property, and it eliminates the need for any other tie-breaking mechanisms.

Predictions are made using the Most Specific Wins (MSW) operator. Given an instance, all applicable rules are identified, and any rule strictly overridden by a more specific applicable rule is discarded. If the remaining rules all agree on a label, that label is the prediction. If no rules apply, the classifier abstains. This mechanism makes the interaction between defaults and exceptions explicit, but it relies on the assumption that specificity alone is sufficient to resolve all conflicts. The full chain of defaults and overrides is visible in the ruleset itself, which produces shorter and more stable explanations.

As an example, consider a scholarship policy represented as an ECCRS ruleset [13]. Let the binary features in a dataset be:

Atom	Meaning
a_1	high GPA
a_2	financial need
a_3	community service
a_4	disciplinary flag
a_5	research project

A default rule might say that a high GPA warrants the award. A more specific rule adds that a disciplinary flag overrides this. A

yet more specific rule adds that a research project overrides the disciplinary flag, reinstating the award. A more specific rule says that if you have financial need and do community service, you warrant the award, and a final most specific rule overrides this if you also have a disciplinary flag. This example of a rule set looks like:

Rule	Body	Label
r_1	a_1	1 (award)
r_2	$a_1 \wedge a_4$	0 (deny)
r_3	$a_1 \wedge a_4 \wedge a_5$	1 (award)
r_4	$a_2 \wedge a_3$	1 (award)
r_5	$a_2 \wedge a_3 \wedge a_4$	0 (deny)

For any student instance, the prediction follows this chain. The most specific applicable rule wins, and the explanation is simply the sequence of rules that applied and which ones overrode which.

ECCRS supports two different forms of the closure property. Relaxed closure requires that conflicting rules are structurally comparable only within observed training instances. Strict closure extends this to the entire Boolean feature space including unseen feature combinations, and is the stronger condition required for alignment with formal defeasible reasoning frameworks [13].

While ECCRS provides a clean structural treatment of defaults and exceptions, its behaviour and scalability have not yet been validated. We rely on the explanation provided by Madzime [13]. As ECCRS is under active development at UCT, this project contributes to its validation by examining how its conclusions align with established KLM closure methods. This project investigates the translation between ECCRS and KLM for the purpose of verification under three closure methods: Rational Closure, Lexicographic Closure, and System W. Each of these captures default-and-exception reasoning with varying degrees of permissiveness. No single closure method is universally appropriate, and this project will explore how each one interacts with ECCRS conclusions.

3 Defeasible and Non-Monotonic Reasoning

3.1 Classical Logic and Its Limitations

Classical logic has been the foundation of formal reasoning with regard to AI, but with it comes a major limitation: monotonicity. Once a conclusion follows from a set of statements, it continues to hold no matter what new information is added [8]. This works in mathematics, but breaks down when rules have exceptions.

The classic example is the Tweety example [8, 10]: we know birds fly, penguins are birds, and penguins don't fly. Classical logic leads to a contradiction where a penguin both flies and doesn't fly because it has no way to deal with the rule that says "penguins don't fly" as an exception that overrides the general "birds do fly" rule. The only conclusion is that penguins don't exist, which is unacceptable.

Early attempts to address this, such as Reiter's Default Logic [15], introduced ways of generating tentative, retractable conclusions. These approaches lacked a general framework for comparing and classifying different forms of non-monotonic inference. This led to the development of the KLM framework, which will be introduced in the following sections.

3.2 Non-Monotonic Reasoning and Knowledge Bases

The solution to the limitations of classical logic is to allow conclusions to be retracted when new information is introduced. Rather than saying that birds always fly, we say that birds typically fly. This is a claim that generally does hold, but can be overridden by more specific information. This type of reasoning is called defeasible reasoning, and it aligns more closely than classical logic with how people actually think [10].

Lehmann and Magidor [12] define this using conditional assertions of the form $\alpha \mid\sim \beta$, which we read as "if α then typically β". A set of such assertions is called a conditional knowledge base, and represents everything one defeasibly knows about how the described world behaves in general. In the bird example, the knowledge base would contain $bird \mid\sim flies$ and $penguin \mid\sim \neg flies$. The penguin rule doesn't contradict the bird rule, but it overrides it in the specific case where the bird we are concerned with is a penguin. The knowledge base would look something like this:

$$bird \mid\sim flies$$
$$penguin \mid\sim \neg flies$$
$$penguin \mid\sim bird$$

Kraus et al. [10] introduce a foundation through preferential models, which impose a plausibility ordering over possible worlds. Conclusions are drawn from the most plausible situations that align with the available information, so more specific knowledge takes priority over more general defaults.

Several other frameworks for non-monotonic reasoning were developed around the same time, including Default Logic [15] and circumscription. Kaliski [8] examines these alternatives and notes that while each provides something useful, the KLM framework stands out for its generality and clean semantic theory. This project therefore builds on the KLM framework. We introduce it in the following section.

3.3 The KLM Framework

The KLM framework, introduced by Kraus, Lehmann, and Magidor [10], provides the formal foundation for the defeasible reasoning approach used in this project. Rather than proposing a single reasoning system, KLM identifies families of non-monotonic consequence relations and characterises them by the properties they satisfy, providing a general framework within which different defeasible reasoning systems can be classified.

The semantic core of the framework is the preferential model. A preferential model imposes an ordering over possible worlds according to their plausibility. A defeasible conditional $\alpha \mid\sim \beta$ holds in a preferential model if all the most plausible worlds satisfying α also satisfy β. This confirms what we expect, that defaults hold in typical situations. A bird flies in most normal bird-worlds, even if there exist less typical bird-worlds where it does not [10].

Lehmann and Magidor [12] refine this further by introducing ranked preferential models, where each world is assigned a rank representing how typical or exceptional it is. Lower ranks correspond to more normal situations, and higher ranks to more exceptional ones. This structure makes it possible to easily resolve conflicts between defaults and exceptions. These semantics provide a clean

account of default reasoning, but it abstracts away from how rankings are constructed in practice, which is deferred to specific closure methods.

For this project, the KLM framework operates over propositional logic, in keeping with the binary feature space of ECCRS. An ECCRS ruleset is translated into a KLM conditional knowledge base, and classification queries are answered using closure methods defined over that knowledge base. We consider three methods, namely Rational Closure, Lexicographic Closure, and System W. Each extends the KLM framework with a different strategy for resolving conflicts between defaults and exceptions. These methods are introduced in the following sections.

3.4 Rational Closure

Rational Closure, introduced by Lehmann and Magidor [12], is the most widely used entailment method in the KLM framework and the starting point for this project. The central idea is to rank the defeasible conditionals in a knowledge base by how exceptional they are, and then use those ranks to answer queries by temporarily setting aside the most general information when it conflicts with more specific knowledge.

A formula α is considered *exceptional* for a knowledge base K if its negation is a typical consequence of K that is, $\neg\alpha$ holds in all the most plausible worlds of K. Intuitively, if α describes something that is already atypical given what we know, then any rule whose antecedent is α is itself an exceptional rule. This idea is used to assign every conditional in the knowledge base a rank, which is the lowest rank at which its antecedent is no longer exceptional. Conditionals with lower base ranks are more general and more defeasible, while those with higher ranks are more specific and exceptional. Conditionals whose antecedents are exceptional at every level receive an infinite rank, meaning they behave as classical facts, always the case.

The ranks are computed by the BaseRank algorithm, which works iteratively. Starting from the full knowledge base, it repeatedly removes all conditionals whose antecedents are exceptional at the current level, producing increasingly smaller subsets. Each removed layer is assigned the next rank. The process ends when no further conditionals can be removed, and whatever remains at that point is put into the infinite rank.

As an example, consider the following knowledge base K [8]:

$$\begin{aligned} \mathit{boat} &\mid\sim \mathit{floats} \\ \mathit{leaky} &\mid\sim \mathit{boat} \\ \mathit{leaky} &\mid\sim \neg\mathit{floats} \\ \mathit{FlyingDutchman} &\mid\sim \mathit{boat} \\ \mathit{FlyingDutchman} &\mid\sim \mathit{leaky} \end{aligned}$$

Before applying BaseRank, the knowledge base is converted to its materialisation $\overrightarrow{K}$, replacing each defeasible conditional $\alpha \mid\sim \beta$ with its classical version $\alpha \rightarrow \beta$. The algorithm then operates on these classical implications, and the resulting ranks are inherited by the original defeasible conditionals.

Applying the algorithm, we first check which antecedents are exceptional with respect to the full knowledge base. Since *leaky* and *FlyingDutchman* are both exceptional because their antecedents are not satisfied in any of the possible models of the knowledge base, their conditionals are exceptional. $\mathit{boat} \mid\sim \mathit{floats}$ is not exceptional, so it is assigned rank 0 and removed from consideration. This gives us the following after the first iteration:

Rank	Conditional
0	$\mathit{boat} \rightarrow \mathit{floats}$

The remaining conditionals are checked again. At this level, *FlyingDutchman* is still exceptional but *leaky* is no longer exceptional, so the two leaky conditionals are assigned rank 1 and removed:

Rank	Conditional
0	$\mathit{boat} \rightarrow \mathit{floats}$
1	$\mathit{leaky} \rightarrow \mathit{boat}$
1	$\mathit{leaky} \rightarrow \neg\mathit{floats}$

The two *FlyingDutchman* conditionals remain. At this point no antecedents are exceptional for the remaining subset, so they receive rank ∞, meaning they behave as classical facts:

Rank	Conditional
0	$\mathit{boat} \rightarrow \mathit{floats}$
1	$\mathit{leaky} \rightarrow \mathit{boat}$
1	$\mathit{leaky} \rightarrow \neg\mathit{floats}$
∞	$\mathit{FlyingDutchman} \rightarrow \mathit{boat}$
∞	$\mathit{FlyingDutchman} \rightarrow \mathit{leaky}$

Now consider the query: does a Flying Dutchman typically not float, that is, does $K \models_{RC} \mathit{FlyingDutchman} \mid\sim \neg\mathit{floats}$ hold? The Rational Closure algorithm answers this as follows. It first checks whether the query antecedent *FlyingDutchman* is exceptional with respect to the full ranked knowledge base. It is, because the materialisation classically entails $\neg\mathit{FlyingDutchman}$. So the algorithm removes the rank 0 conditionals and checks again using only ranks 1 and ∞:

Rank	Conditional
1	$\mathit{leaky} \rightarrow \mathit{boat}$
1	$\mathit{leaky} \rightarrow \neg\mathit{floats}$
∞	$\mathit{FlyingDutchman} \rightarrow \mathit{boat}$
∞	$\mathit{FlyingDutchman} \rightarrow \mathit{leaky}$

Is *FlyingDutchman* still exceptional with respect to this reduced knowledge base? No, there now exists a world where *FlyingDutchman* is true without contradiction. So the algorithm stops removing conditionals and checks the query against the current knowledge base. Since $\mathit{FlyingDutchman} \rightarrow \mathit{leaky}$ and $\mathit{leaky} \rightarrow \neg\mathit{floats}$ are both present, the knowledge base classically entails $\mathit{FlyingDutchman} \rightarrow \neg\mathit{floats}$, and so the query is confirmed: a Flying Dutchman typically does not float.

A formal entailment condition can be stated concisely: $K \models_{RC} \alpha \mid\sim \beta$ holds if and only if $\mathrm{br}_K(\alpha) < \mathrm{br}_K(\alpha \wedge \neg\beta)$, or α has infinite rank [12]. In plain terms, the query is entailed when knowing α puts us at a lower rank than knowing α together with the negation of β. This would mean that β is what we would typically expect given α.

Rational Closure is computationally well-behaved, but it has a known limitation. It does not allow exceptional subclasses to inherit generic properties from their superclass [12]. In the penguin example, knowing that penguins don't fly causes Rational Closure to rank penguins as exceptional birds, which can prevent it from concluding that penguins have wings even though there is no reason to think otherwise. This limitation motivates the stronger closure methods described below.

3.5 Lexicographic Closure

Lexicographic Closure, introduced by Lehmann [11], extends Rational Closure to address its inheritance problem. The key insight is that Rational Closure is too conservative when handling exceptional subclasses. By discarding all conditionals of lowest rank when an antecedent is exceptional, it throws away too much information, including generic properties that should still be inherited.

Lexicographic Closure addresses this by introducing a more refined preference ordering over possible worlds. Rather than simply discarding conditionals by rank, it orders worlds according to how seriously they violate the knowledge base. Violating a more specific conditional is considered more serious than violating a less specific one, and violating more conditionals at the same level of specificity is more serious than violating fewer. These two criteria are combined lexicographically. Specificity is the primary attribute we look at, and cardinality is the secondary one. The result is a much richer ranking of worlds, which allows Lexicographic Closure to retain more information when resolving queries about exceptional subclasses.

Lehmann identifies four principles that contribute to sound default reasoning [11]: the presumption of typicality, which says we should assume typical behaviour unless there is a reason not to; the presumption of independence, which says we should assume that unrelated features do not affect each other; priority to typicality, which resolves conflicts between the first two in favour of typicality; and respect for specificity, which says that more specific rules take priority over more general ones. Lexicographic Closure is designed to satisfy all four.

To see the practical difference from Rational Closure, consider a knowledge base K containing $boat \mathrel{|\!\sim} floats$, $boat \mathrel{|\!\sim} sailors$, and $leaky \mathrel{|\!\sim} \neg floats$. Rational Closure cannot conclude that $leaky \wedge boat \mathrel{|\!\sim} sailors$, because when handling the exceptional leaky boat case it discards the boat conditionals, losing the sailors information along with the floats information. Lexicographic Closure correctly concludes $leaky \wedge boat \mathrel{|\!\sim} sailors$, because violating $boat \mathrel{|\!\sim} sailors$ is treated as more serious than violating $boat \mathrel{|\!\sim} floats$ in the context of a leaky boat, and so the sailors conditional is retained [11].

Lehmann shows that Lexicographic Closure extends Rational Closure. Every inference derived by Rational Closure is also derived by Lexicographic Closure, but not vice versa [11]. It is therefore a strictly richer entailment relation, and a natural next step when Rational Closure proves too conservative for a circumstance.

3.6 System W

The drowning problem refers to the failure of a reasoning system to infer properties of a superclass for a subclass that is exceptional only in some unrelated respect. For example, knowing that penguins do not fly should not prevent us from concluding that penguins have wings. Penguins are exceptional birds with respect to flying, but not with respect to having wings.

System W, introduced by Komo and Beierle [9], is the most permissive of the three closure methods considered in this project. It strictly extends both Rational Closure and c-inference (another closure method), which is a ranking-based entailment method that takes all compatible ranking models into account. The key idea behind System W is to order possible worlds according to which conditionals they falsify. A world is preferred over another if it falsifies a less serious set of conditionals, where seriousness is determined by the tolerance partition of the knowledge base. A tolerance partitioning is a partitioning of conditionals into layers based on which ones can be consistently assumed to hold together. This gives System W a preference ordering that allows it to make distinctions that Rational Closure and Lexicographic Closure cannot.

As an example, consider the following knowledge base K [9]:

$$\begin{aligned} bird &\mathrel{|\!\sim} flies \\ penguin &\mathrel{|\!\sim} \neg flies \\ penguin &\mathrel{|\!\sim} bird \\ bird &\mathrel{|\!\sim} wings \\ flies &\mathrel{|\!\sim} airborne \end{aligned}$$

Rational Closure fails to conclude $penguin \mathrel{|\!\sim} wings$. The reason is that penguins are ranked as exceptional birds, and this exceptionality contaminates all bird properties, including having wings, even though there is no conflict between penguins and wings in the knowledge base. System W avoids this by reasoning about which conditionals each world actually falsifies. A world where a penguin has wings only falsifies the flying conditional, which belongs to a lower-priority partition layer. A world where a penguin lacks wings falsifies both the flying conditional and the wings conditional, which is strictly more serious. System W therefore prefers the world where the penguin has wings, and correctly concludes $penguin \mathrel{|\!\sim} wings$ [9].

As Kaliski [8] observes, no single closure method is universally appropriate, and the choice between Rational Closure, Lexicographic Closure, and System W is dependent on context. System W provides the richest set of entailments, but this comes at the cost of greater complexity and practical usability. Understanding which closure method produces the most appropriate entailments for a given ECCRS ruleset is a central question this project considers.

4 Bridging ECCRS and Defeasible Reasoning

4.1 Translating ECCRS into KLM Conditionals

The first step in the verification workbench is to translate an ECCRS ruleset into a KLM conditional knowledge base. Darwiche and Hirth [4] show that the input-output behaviour of a classifier can be encoded as a propositional formula, allowing formal reasoning about decisions. This includes computing sufficient and complete reasons for why a particular classification was made. ECCRS makes this translation natural because its rules are structured as defeasible conditionals.

The translation works by introducing a fresh propositional atom m to represent the positive class label, so that m corresponds to label 1 and $\neg m$ to label 0. Each ECCRS rule (B, ℓ) is then mapped to a defeasible conditional in the KLM knowledge base $K(R)$: a rule with label 1 becomes $B \mid\sim m$, and a rule with label 0 becomes $B \mid\sim \neg m$ [13]. Returning to the scholarship example from Section 2.3, the five rules translate directly into the following KLM knowledge base $K(R)$, where m refers to the awarding of the scholarship:

$$\begin{aligned} a_1 &\mid\sim m \\ a_1 \wedge a_4 &\mid\sim \neg m \\ a_1 \wedge a_4 \wedge a_5 &\mid\sim m \\ a_2 \wedge a_3 &\mid\sim m \\ a_2 \wedge a_3 \wedge a_4 &\mid\sim \neg m \end{aligned}$$

The resulting knowledge base contains exactly one conditional per rule, preserving the specificity ordering between rules. Rienstra [17] motivates the use of non-monotonic background knowledge in this setting, showing that sufficient reasons for classifier decisions can be unnecessarily long when background knowledge about relationships between features is ignored, and that preferential models provide a way to fix this.

For this translation to be well-defined, three conditions on the ruleset must hold: strict global exception closure, which ensures no unresolved conflicts exist anywhere in the feature space; sanity, which rules out contradictory rules with identical bodies; and the no total override condition, which ensures every rule has at least one instance where it fires without being overridden [13]. Once this translation has taken place, we need to verify it.

4.2 Verification via Closure Methodologies

Once the ECCRS ruleset has been translated into a KLM knowledge base, classification queries can be answered using defeasible entailment. Given a complete feature pattern F, the workbench queries whether $F \mid\sim m$ or $F \mid\sim \neg m$ is entailed, corresponding to a positive or negative classification respectively.

The central result is the alignment theorem [13]: under the three assumptions above, the answers produced by applying Rational Closure on the translated knowledge base agree exactly with the MSW predictions of the original ECCRS ruleset, including abstaining when no rule applies. While the alignment theorem is stated for Rational Closure, this project extends the verification to Lexicographic Closure and System W.

Chama et al. [3] introduce an algorithm that identifies the minimal subsets of a defeasible knowledge base responsible for a given entailment. Their work establishes that explaining defeasible conclusions requires first establishing the ranking of statements in the knowledge base, since only a portion of the knowledge base is active at any rank level. This ranking-based approach is directly relevant to how the workbench generates explanations, where a conclusion is explained by identifying which ranked conditionals were decisive and which were discarded for being too exceptional.

4.3 Explanation Traces

Explanation traces are the primary mechanism for making ECCRS predictions transparent and verifiable. The workbench produces explanations on two sides simultaneously. On the ECCRS side, the trace lists all applicable rules for a given instance, identifies which rules are overridden by more specific ones, and reports the most specific applicable rules that determine the final label. On the KLM side, the trace reports which closure method entailed the classification query, and records the rank comparison that drove the result.

Chama [2] introduces English explanation templates for defeasible conclusions, showing that explanations can be presented in natural language that users can follow. This relates to the explanation traces produced by our workbench, which aims to present the default-and-exception reasoning in a form that reflects how a human would explain the decision. The scholarship example from Madzime [13] illustrates this. For a student with a high GPA, a disciplinary flag, and a research project, the trace reports that the default award rule was overridden by the disciplinary exception, which was itself overridden by the research project exception, resulting in a final label of award. This two-sided explanation approach connects the formal reasoning of the KLM framework to the human-readable narrative that ECCRS is designed to produce.

5 Discussion/Comparison of Prior Work

The literature reviewed in this paper spans three intersecting areas: explainable AI and symbolic learning, defeasible and non-monotonic reasoning, and the bridge between them that our verification workbench represents.

With regard to symbolic classifiers, the case for transparent models over post-hoc explanation is well established. Rudin [18] argues that a surrogate model is not the same as the model it approximates, and that for consequential decisions this difference is unacceptable. The COMPAS example [6, 18] makes this concrete: a system that is black-box and biased cannot be corrected if its reasoning cannot be examined. CORELS [1] and Boolean formula classifiers [7] demonstrate that transparency does not require sacrificing accuracy, undermining the assumption that interpretability comes at a performance cost. Both approaches are transparent in the sense that their rules can be read, but neither produces explanations that reflect how humans reason. In a rule list, a default and its exception appear as separate ordered rules with no explicit relationship. In a Boolean formula, exceptions are buried in nested logical structures. The explanation is traceable but not natural.

Darwiche and Hirth [4] provide a formal basis for symbolic explanation, introducing complete and sufficient reasons for classifier decisions. Rienstra [17] identifies a practical weakness in this approach, that sufficient reasons can be unnecessarily long when they include features that are technically relevant but redundant given known relationships in the data. Rienstra's proposed solution of extending the explanation system with non-monotonic background knowledge points toward the framework this project employs. It is worth noting that this problem arises not from the classifier being opaque, but from the explanation framework being too flat. Even fully transparent symbolic classifiers can produce explanations that are formally correct but practically uninformative, which motivates the use of defeasible reasoning not just for verification but for explanation quality.

DHCE [14] addresses the naturalness gap by grouping defaults with their exceptions into normally-unless statements, producing step-by-step traces that reflect the structure of human decision-making. Its performance is strong, matching or outperforming other interpretable models on standard benchmarks. The limitation is that DHCE is not grounded in a formally defined non-monotonic framework, which means its explanations cannot be formally verified. Without a formal grounding, there is no principled way to check whether the reasoning DHCE attributes to a decision is the reasoning that a logically consistent framework would support. ECCRS [13] closes this gap by embedding the same design goals within a formal framework, making verification possible.

Default Logic [15] and circumscription were early attempts to formalise defeasible inference, but as Kaliski [8] notes, they lacked a general framework for comparing different consequence relations. KLM [10] fills this gap by characterising families of non-monotonic consequence relations through the properties they satisfy. The ranked preferential models of Lehmann and Magidor [12] give this framework a clean semantic foundation, assigning worlds ranks that reflect their typicality and resolving conflicts by preferring more typical worlds. This is the structure needed to verify ECCRS predictions. The MSW operator and the ranked knowledge base are performing similar operations, both preferring the most specific applicable information and setting aside more general defaults when they conflict. This structural correspondence is what makes the translation well-defined and the alignment theorem non-trivial.

Chama [2] shows that explanations can be produced in natural language, confirming that the formal and human-readable traces are complementary rather than in conflict. This means the verification is not merely a formal check but one that can be communicated to the user in the same terms as the ECCRS trace itself. Comparing the two traces is the mechanism by which the human-readable justification produced by ECCRS is connected to the formal semantics of defeasible reasoning, and it is this connection that prior approaches in this line of work have lacked.

The alignment theorem [13] guarantees that Rational Closure agrees exactly with MSW predictions under the conditions of strict global exception closure, sanity, and no total override. This project extends the verification to Lexicographic Closure and System W, which capture default-and-exception reasoning with greater permissiveness. The central empirical questions this project addresses are whether these methods produce different verdicts on real ECCRS rulesets, and what any divergences reveal about the limits of MSW.

The open question is whether the structural constraints imposed by exception closure already render Lexicographic Closure and System W redundant on any well-formed ECCRS ruleset; answering it requires validation across different classification tasks, which the workbench is designed to support.

6 Conclusion

This review has followed a clear progression from post-hoc explanation methods to formally grounded symbolic classifiers. Surrogate models such as LIME provide interpretability but cannot guarantee that their explanations reflect the model's actual reasoning. Symbolic classifiers like CORELS and Boolean formula classifiers restore that guarantee but cannot express the default-and-exception structure that makes explanations feel natural. DHCE captures that structure but lacks formal grounding. ECCRS addresses all three limitations, embedding exception handling within the KLM framework and enabling verification through defeasible entailment.

The verification workbench translates ECCRS rulesets into KLM conditional knowledge bases and compares the explanation traces produced by the MSW operator against those derived by Rational Closure, Lexicographic Closure, and System W. The alignment theorem guarantees exact agreement between Rational Closure and MSW under the required structural conditions, and the extension to the stronger closure methods provides a basis for determining whether those conditions are already sufficient to make the richer entailment methods redundant. The two-sided explanation trace is the mechanism that makes this verification legible: it connects the formal semantics of defeasible reasoning to the human-readable justifications that ECCRS is designed to produce, providing the kind of correctness guarantee that prior approaches in this line of work have lacked.

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